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Structural Monitoring and Rehabilitation of a Gerber Bridge

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Many existing reinforced concrete bridges exhibit behavior at the Service Limit State and the Ultimate Limit State which can be considered unsatisfactory with respect to the current provisions of Codes, due to deficiencies deriving from obsolete calculation methods, ageing, material degradation, diffuse or localized corrosion and increased loads. Among these, cantilever concrete bridges with half-joints, also called Gerber bridges in Europe, may present a decrease of safety conditions for the most stressed current sections of the deck or for local potential failure mechanism of critical members, that affect the performance of elements sensitive to degradation, such as Gerber saddles. To perform vulnerability assessment, the validation of FEM models against on-site tests and investigation is of fundamental importance for acquiring information on the constraint conditions, verifying the structural scheme by identifying the degree of collaboration between the structural elements and identifying the state of cracking or damage. Thus, for a case of study the results of a structural assessment performed with Finite Element model are presented through validation with on-site tests. Then, a simple strengthening intervention through external prestressing which changes the structural behavior in service life, is implemented without the need to recourse to interventions of greater impact, obtaining an improvement of the overall performance. The results of monitoring and on-site load tests are presented as the validation method for the analyses on different FE models together with the proposed intervention.

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1. Introduction

The structural assessment of existing concrete and prestressed bridges implies an adequate level of knowledge of the structure and of its design and construction history. For the assessment of the state of stress in service, it is also necessary to carry out reliable Finite Element numerical models that provide stress and deformation under loads for the check of structural elements and for the evaluation of deficit which may be detected. The actual construction of the bridge can deviate significantly from the provisions of these models due to many factors: the real constraint conditions, the stiffness of the cracked elements, the transverse behavior of the multi-beam side by side decks, the construction phases that determined the configuration for permanent loads, as well as any degradation for localized or diffuse corrosion.

The degradation or damage of a bridge often penalizes the performance at Serviceability or Ultimate Limit State, with temporary traffic restrictions, pending maintenance. In these situations, bridge monitoring is generally the option chosen by several Road Authorities for the assessment of safety level ensured by the bridge. Among the monitoring methods for assessment, load testing with different load configurations plays an important role. The recent Guidelines of the Italian Code on the bridges (Italian Ministry of Infrastructures, 2020), in fact, consider decreasing levels of traffic loads for the evaluations of transit ability and operativity. The adoption of a load configuration on the bridge and the relative load test, allow engineers to know the response of the bridge (Zarate Garnica et al., 2022). Displacement and strain measurements under load provide fundamental information for the validation of numerical models because the deformation response lets the model calibration both in terms of the stiffness of structural elements and in terms of the bearing functionality. The on-site investigation of materials, degradation status, bearing type, provides together with the load test, a representative depiction of the bridge that can be fully transferred into the numerical model to obtain more reliable results of the analyses carried out.

Lantsoght et al. (2017) provide a useful state-of-the-art on load testing of concrete bridges, in which techniques for load application, measurements and procedures are described, just in view of the assessment of existing bridges. Aguilar et al. (2015) uses the load test as a method for estimation of state of stress in a prestressed concrete bridge made of adjacent beams in a classical grid scheme. Abedin et al. (2022) applied the load testing with different measurement procedures for implementing a model updating methodology in monitoring and assessment of existing bridges.

In this paper a case-study of a bridge built by Riccardo Morandi in Southern Italy (Lo Giudice et al., 2016; Granata et al., 2022) is presented, which was monitored by load testing for the implementation of on-site results in numerical FE models in the view of carrying out reliable analyses on these models, validated in the service state by the results of load tests. Particularly the aspects of concrete elements stiffness in longitudinal and transverse directions (Manterola, 2006) are highlighted together with the role of bearings. Afterwards a brief presentation of the rehabilitation hypothesis carried out with external prestressing is given, as a result of the structural assessment.

2. Structural monitoring for safety assessment

2.1. Case-study bridge

The bridge is placed over the Salso river in Licata (Sicily, Southern Italy) and it was designed by Riccardo Morandi; it is a cantilever Niagara-type bridge with three spans (also called Gerber bridge), with side spans of 33.10 m and a central span, composed of two cantilevers and a simply supported central beam (drop-in span), with a total length of 49.6 m. The overall width of the deck is 19.00 m, while the total length is 115.80 m. The bridge is shown in Figure 1.

Transversely, the two side spans and the cantilevers are composed of eight side-by-side T-beams with a web thickness of 400 mm, variable height from 1.40 m to 2.50 m and upper slab 200 mm thick. The girder has three transverse beams for each span and a bottom slab with thickness varying from 220 to 300 mm near the piers. The drop-in span is made of eight prestressed reinforced concrete beams with variable height and double T section, completed by a reinforced concrete slab with a thickness of 200 mm and three transverse beams. The central beams are prestressed through the technique patented by Morandi of M5 type tendons (Morandi, 1970). The geometry, design values of prestressing forces and original loads were obtained from the original documents and confirmed by on-site investigations with limited scattering. Materials are concrete for side spans and cantilever with compressive strength

$f_{ck} = 25$ MPa; concrete for prestressed girder with $f_{ck} = 45$ MPa; ordinary reinforcement steel type Aq50, with yielding strength $f_{yk} = 270$ MPa; prestressing steel with tensile failure strength $f_{ptk} = 1700$ MPa.



Fig. 1. Case-study bridge. a) General view. b) View of Gerber saddle with cantilever and drop-in span

2.2. Monitoring through on-site load tests

The bridge was monitored through several configurations of load tests by comparing the actual load applied on the bridge by trucks with those required by Italian Standards and Eurocode 1 (CEN, 2005), through the use of the influence lines of the most significant cross-sections for maximum shear, bending moment and displacement effects. Figure 2 shows the transverse configuration of Code moving loads according to the Courbon-Engesser theory for maximizing the effects in terms of shear and bending on the outermost beam.

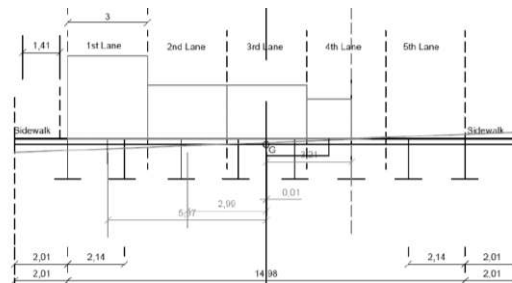


Fig. 2. Transverse distribution of moving loads with lanes for Courbon-Engesser theory to maximize the effect on the outermost beam.

Figure 3 shows phases of the load test with trucks.



Fig. 3. Views of load test phases.

The main load schemes used for the drop-in span and side spans are reported by Figure 4. Measurement was carried out with a double method: inclinometers placed on the pavement surface at the outermost beam together with topographical control measurements by a total station. It is worth noting that it was not possible to make direct measurements with instrumentation from below because of the river under the bridge, so the measurements were all conducted from the deck surface. Zarate Garnica et al. (2022) reports this measurement methodology as "very sensitive" and among the most widely used techniques. It should be noted, however, that in this case the use of the inclinometers must be suitably modified by the placement of two instruments immediately before and after the Gerber saddles, to measure the relative rotation in each load phase that occurs between the cantilever and the drop-in span.

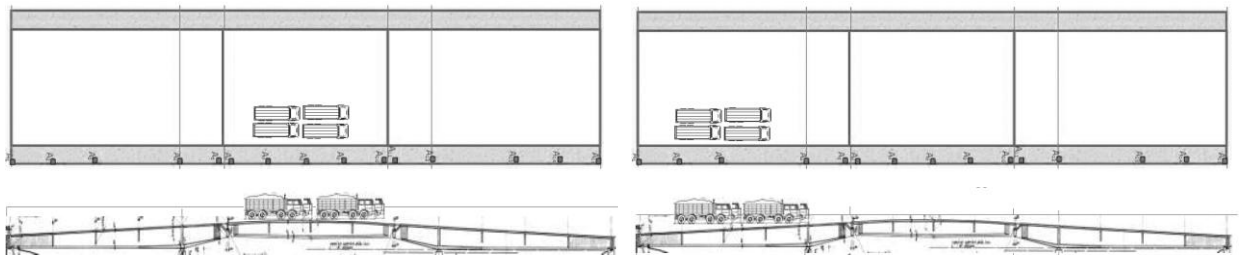


Fig. 4. Main load schemes with trucks on the bridge. a) Symmetrical configuration with trucks on the drop-in span. b) Asymmetrical configuration with trucks on the side span.

2.3. FE models, validation and updating.

For the structural assessment of the bridge two different FE models (Figure 5) were considered:

- 1) The first model (Model A) is made of beam elements only for longitudinal main girders and crossbeams; it has 298 nodes and 398 frame elements. External restraints are modelled as rigid restraints on piers and abutments. Bearing restraints are arranged symmetrically with fixed bearings on piers and moving bearings on abutments. The internal restraints on Gerber saddles are modelled through beam releases to bending moments on the two ends of the drop-in span and through release to axial force on one end only. The application of moving loads is carried out directly on the beams through the transverse distribution of Courbon. This FE model is the closest to Morandi's original design calculation by hand.
- 2) The second model (Model B) is made of a lattice of longitudinal and transverse beams with the addition of shell elements on the extrados which simulate the presence of the deck slab. This model takes into account the stiffness of the upper slab both in longitudinal and transverse directions, for a better transverse redistribution of moving loads. This is a model which can be considered closer to the physical reality at SLS in which the slab collaborates with the beams in the side-by-side behavior, improving the result in terms of load deflections. This model presents 622 nodes, 601 frame elements and 504 shell elements, assembled according to a mesh that takes into account the connection points between the longitudinal and transverse beams, to guarantee the conditions of compatibility between elements.

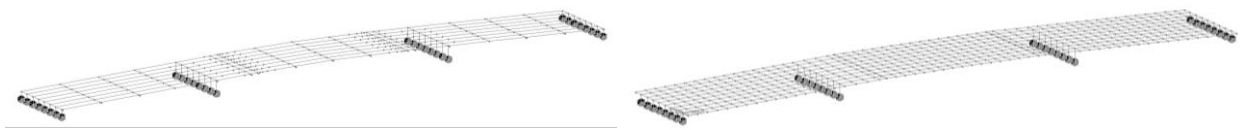


Fig. 5. FE models for numerical analyses. a) Model A. b) Model B

The validation of the above models was carried out through the measurements made during load testing and after on-site investigations on the materials (concrete and steel reinforcing bars) and on the distribution of reinforcements, comparing the results with original drawings and design specifications. The deflections obtained from the load tests were compared with the results obtained from the two FE models allowing to calibrate the degree of internal constraint

between transverse and longitudinal beams and the effectiveness of the transverse redistribution of the grid system, almost in the elastic phase.

Figure 6 shows a comparison between deflections in the outermost beam of load tests and those supplied by the FE models for two cases of symmetrical truck configurations with two pairs of adjacent trucks (Fig. 6a) and four in-line trucks (Fig. 6b) on the loaded drop-in span.

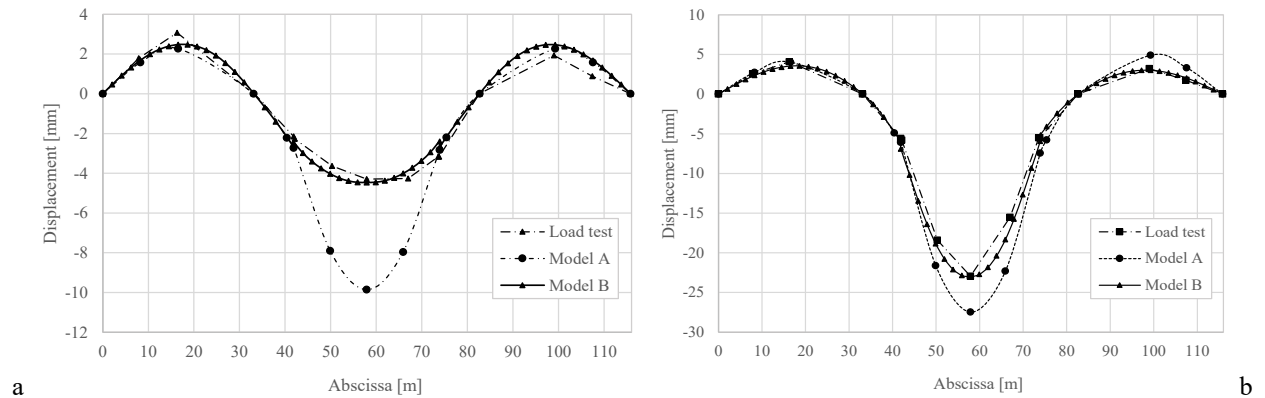


Fig. 6. Symmetric loads on the drop-in span (a) transverse distribution with two pairs of adjacent trucks; (b) transverse distribution with in-line 4 trucks on the outermost beam.

It is worth noting that, since the applied loads are compatible with the Serviceability, the response given by model B is closer to the actual behavior of the bridge, while those provided by Model A show a worst agreement with the results of the load tests, corresponding to a theoretical increase in deformability. This is expected because in linear elastic field many authors highlighted the difference between numerical models and stiffness ratios of grid systems and the methodology to achieve a reliable result in terms of internal forces or displacements (Manterola, 2006). Hence, the validation of Model B is considered more satisfactory for the structural assessment and for a more realistic evaluation of the transverse behavior of the girder at SLS.

The same was done with asymmetric loads (Fig. 7) for the two configurations of trucks on the side span.

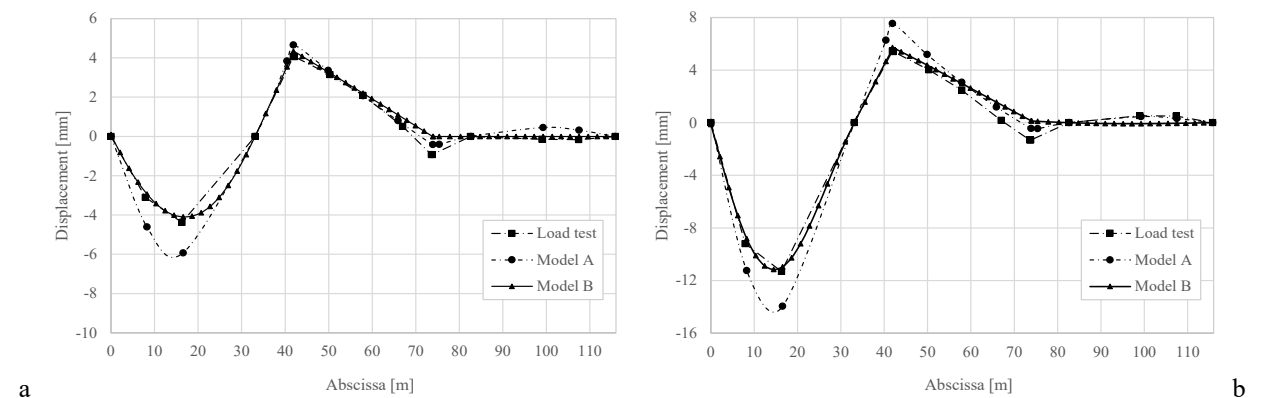


Fig. 7. Asymmetric loads on the side span (a) transverse distribution with two pairs of adjacent trucks; (b) transverse distribution with in-line 4 trucks on the outermost beam.

The results shown in Figures 6 and 7 for both models were obtained by calibrating the degree of interaction between the transverse and longitudinal beams of the central and side spans, that is by varying the effective width of the slab in the definition of the crossbeam properties and then, the effective inertia and stiffness of these elements, together with the difference of Elastic modulus for different casts of concrete.

In addition, another element that greatly affects the overall deformability of the bridge under loads is the degree of the constraint within the Gerber saddle, since the central beams could not always be completely free to rotate, i.e. the relative rotation between the end of the cantilever and that of the drop-in span may or may not be completely free within the joint. Direct on-site tests, endoscopes and Ground Penetrating Radar tests can assess the internal conditions of saddles and bearing supports. In the present case, a free saddle has been considered for rotations because very slight friction occurs between the contact surfaces of the lower saddle corbel and the dapped-end beam of the central span with limited resistance to the opening of the joint when deflection of spans occurred. The saddles of the outermost beams are partially occluded probably due to a maintenance intervention, but the obstruction of the saddle does not seem to compromise its rotational capacity significantly. This issue was taken into account in the FE models, by reducing the effectiveness of the horizontal moving bearing on one saddle, i.e. limiting the degree of the sliding constraint and consequently obtaining better agreement between the deflections measured during the in-situ tests and those provided by the models.

3. Rehabilitation

3.1. Results of structural assessment

The results of the analyses carried out with the above two models in terms of internal forces and the checks of the most stressed cross-sections provided a frame of the assessment against gravitational loads and maximum moving loads. An accurate evaluation was also carried out with detail models on the Gerber saddle (Granata et al., 2023). It has been found that the saddle in undamaged condition is not to be considered, in the specific case, a critical point but the sections of the concrete cantilever with ordinary reinforcements are to be considered more vulnerable than the saddle and the drop-in span that is prestressed. In fact, the simultaneous presence of high values of shear force and negative bending moment in the cantilever cross-section above the pier (Granata et al., 2021), for the maximum moving loads, led to obtain the minimum safety factor where the strong interaction between shear and bending modifies the Ultimate Limit State behavior of the cantilever cross-sections (Granata et al., 2022).

In the case of saddle damaged by corrosion, an increase in damage compared to the situation detected on-site, would bring the saddle to a lower safety factor, becoming a critical section like that of the cantilever. Thus, among the different possible strengthening strategies (Colajanni et al., 2017; Spinella et al., 2019) it seems more reasonable to eliminate the cause of the degradation, eliminating the joint and then the saddles and changing the static scheme of the bridge into a continuous girder.

The most important consequence of the scheme change from Gerber to continuous beam is the rise of bending moments in the saddle areas. The bending moment value remains almost zero for permanent loads, but positive and negative significant values occur in the cantilever and saddle Gerber sections in the continuous beam, as well as negative moments in the whole area of the central beam. Figure 8 shows the diagrams of maximum and minimum moving loads for the Gerber and continuous girders.

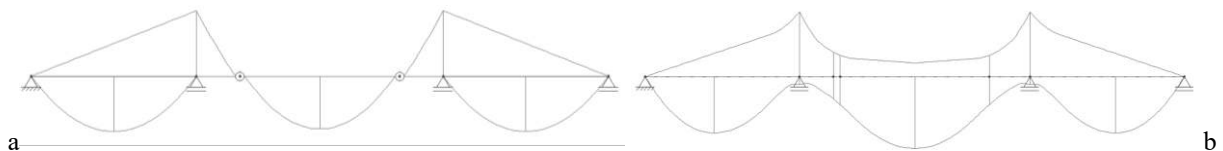


Fig. 8. Max and min bending moment diagram for moving loads. a) Gerber girder. b) Continuous girder

3.2. Rehabilitation through additional external prestressing

From the results of the structural assessment, the most advisable intervention is to eliminate the causes of degradation by closing the saddles Gerber and changing the static scheme of the bridge, improving the Ultimate Limit State safety condition of cantilever sections and addressing the upper stresses in drop-in span that are not present in the original scheme but appear in the continuous beam for moving loads.

This can be done through the application of external prestressing (Recupero & Granata, 2013; Recupero et al., 2014) with the following effects: 1) prestressing is able to deal with the positive and negative bending moments that arise in the Gerber saddle, after closing it; 2) it is able to increase the strength to maximum negative bending moments of the bridge on piers, thus constituting a local strengthening of cantilever sections; 3) it is able to increase the combined shear-flexure strength of the cross-sections of the central beam, initially isostatic and already prestressed by internal tendons with a classical parabolic layout (originally designed for a simply supported beam).

Figure 9 shows the longitudinal profile of the bridge with the external tendons.

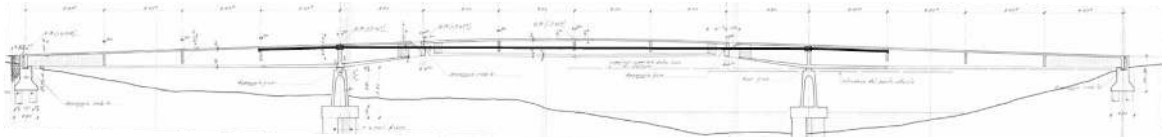


Fig. 9. External prestressing for rehabilitation of the bridge

The most effective configuration of tendons is the rectilinear one that avoids local effects along the bridge due to the deviation of the tendon. Then, a nearly centered configuration that has a sealing function for Gerber saddles without the need to introduce large quantities of ordinary reinforcement inside the saddle was considered the best solution. The tendon maintains a small upper eccentricity near the pier and a slight lower eccentricity in the central span, thanks to the variable longitudinal section of the beam and the original camber.

To make the prestressing effective, it will be necessary to modify the configuration of the bearings because in the original configuration fixed symmetrical supports on the piers and mobile supports on the abutments were placed. This is not suitable for the application of prestressing, and it is not favorable from the seismic point of view, as it induces concentrated horizontal forces on the piers. It was therefore decided to make the end support fixed on one abutment by inserting a seismic damper and to make all the other supports sliding, minimizing friction and making it easy to transfer the prestressing force to the girder.

3.3. Comparison of the structural behavior under maximum service loads before and after the rehabilitation

The consequences in terms of deformability due to the change of the static scheme can be evaluated through the model B, considering the modification of the bearing supports and comparing the behavior of the Gerber bridge with that of the continuous girder.

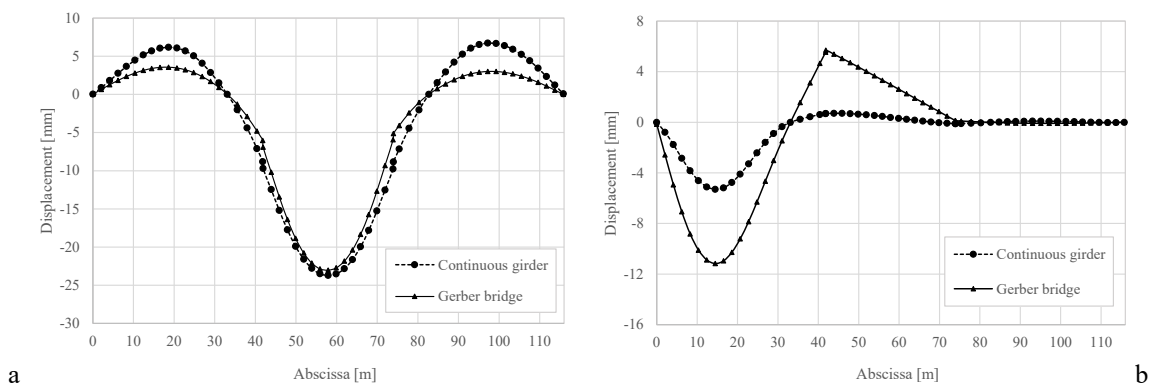


Fig. 10. Comparison between deflections for the test loads in the Gerber bridge and in the continuous girder. a) Symmetrical loads on the drop-in span. b) Asymmetrical loads on the side span

Figure 10 shows the comparison in terms of deflection for the symmetrical loads on the central span and for those on the side span. It can be observed how the continuity due to the closure of the Gerber saddle leaves almost unchanged

the deflection in the midspan while slightly increases the corresponding upward displacement on the side spans, due to the effect of the longitudinal displacement of the restraints on the piers. On the other hand, asymmetric loads have a lower deformability effect with a strong reduction in deflection and therefore a beneficial effect of overall stiffening.

4. Conclusions

The simplest and most available monitoring method for the structural assessment of an existing bridge is to carry out load tests under different load conditions that simulate the loads provided by Codes. This evaluation, although carried out in Serviceability, gives useful information for the construction of reliable numerical models for the subsequent assessments at the Ultimate Limit State or for successive non-linear static analyses on the bridge. The validation of numerical models through load tests provides a tool that permits to update the FE model with information from on-site investigations, in order to have a live model where the main variables can be introduced that can have a significant influence on the results of the assessment. Among these, there is of course also the degradation due to the ageing of the structure and the corrosion of reinforcements. A case-study of a three-span Gerber bridge was presented in this regard, for which the structural assessment procedure was performed based on numerical models validated through on-site investigations and load tests. Symmetric and asymmetric load configurations provide information on the different elements of the bridge and the longitudinal and transverse stiffness ratios of these elements for the different areas of the deck. The results show the usefulness of the loading tests and indications on the most appropriate structural rehabilitation intervention to the problems encountered. As strengthening intervention, a change of the static scheme was chosen, through the use of external prestressing. The results in terms of deflections on the continuous bridge compared to the Gerber one was discussed, for an evaluation of the post-intervention structural behavior. New load tests on the modified bridge after rehabilitation could be useful for evaluating the effectiveness of the intervention or compliance with the design forecasts of the structural retrofit.

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